

STAD-HD: Spatial Temporal Anomaly Detection for Heterogeneous Data through Visual Analytics

Solution for 2016 VAST Challenge MC2 & MC3

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Outline

- Data description
- System introduction
- Cases
- Conclusion

Data description

- GAStech company moved to a three-storey building
  - 125 employees
  - 41 energy zones vs. 23 prox zones
  - 14 days’ static data + 60 hours’ streaming data

Data description

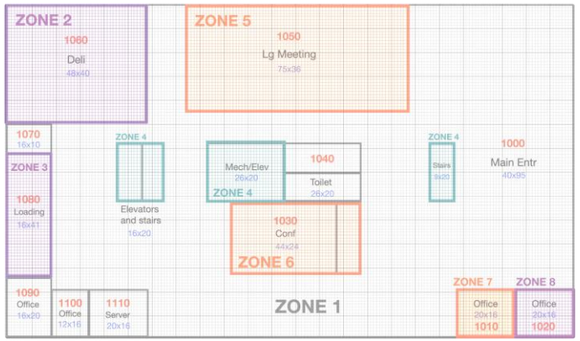
- Building data
  - Energy zone (≤12 attributes)
  - Floor (≤11 attributes)
  - Building (16 attributes)
- Prox data
  - Mobile sensor
  - Fixed sensors

Data description

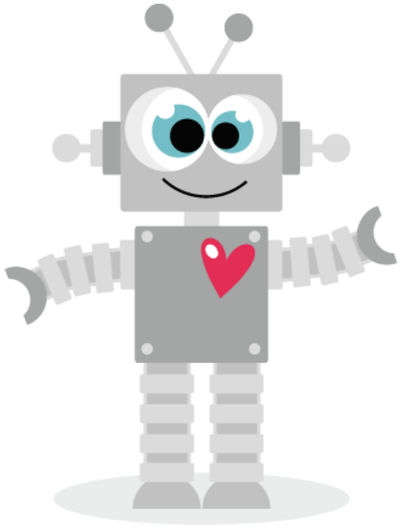
- Prox data



Prox card



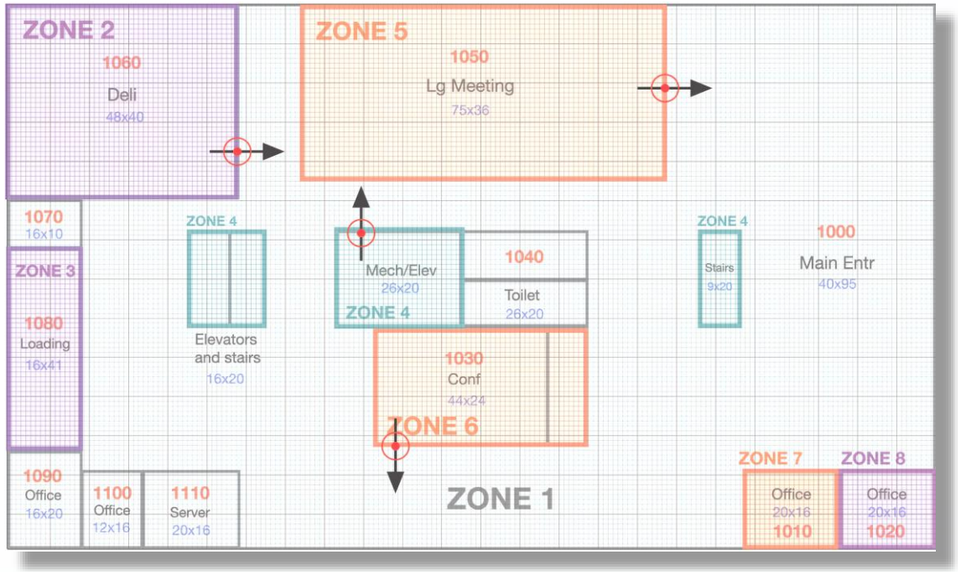
Prox zone



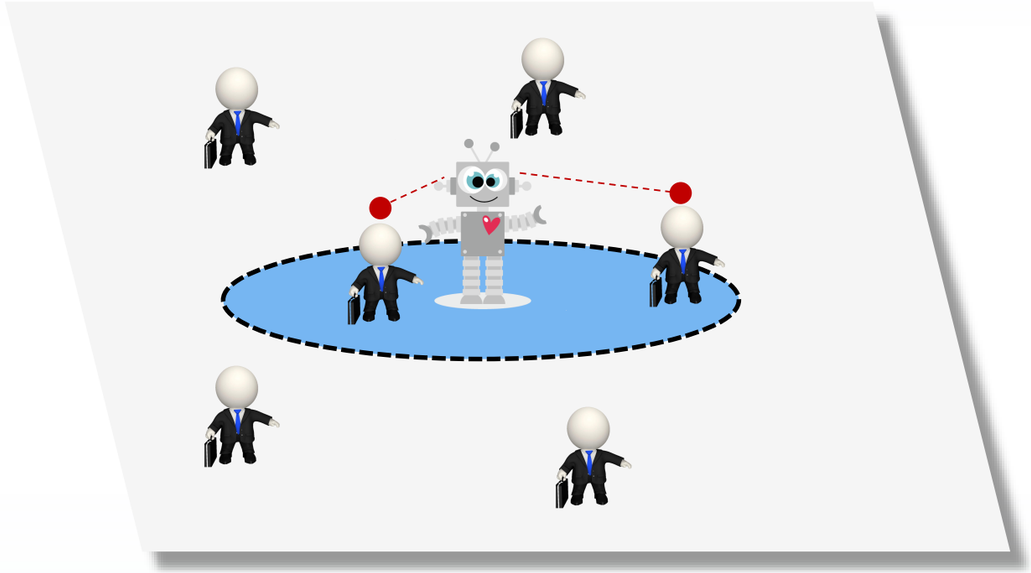
Robot

Data description

- Prox data



Prox-zone detection
Time: Accurate
Position: Inaccurate



Robot detection
Time: Inaccurate
Position: Accurate

Task

- Typical patterns
- Notable anomalies
- Relationships between two types of data

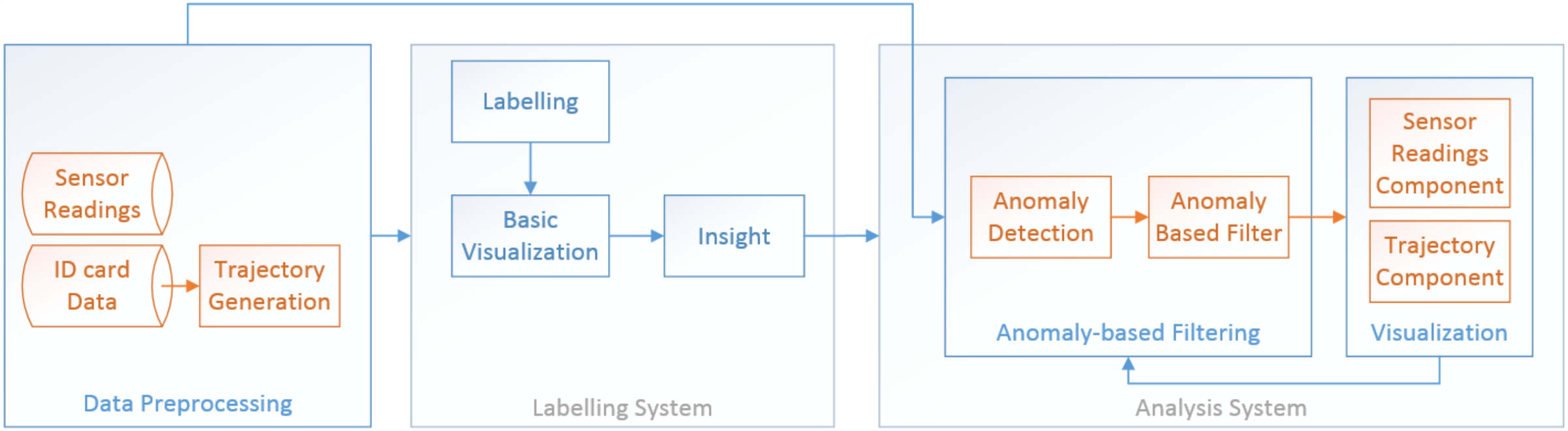
Task

- Typical patterns
- Notable anomalies
- Relationships between two types of data
  - Design requirement: spatial and temporal filters

Task

- “Pattern”
- “Anomaly”
- Not well-defined

Work flow


Sensor Readings
ID card Data
Trajectory Generation
Data Preprocessing
Labelling
Basic Visualization
Insight
Labelling System
Anomaly Detection
Anomaly Based Filter
Anomaly-based Filtering
Sensor Readings Component
Trajectory Component
Visualization
Analysis System

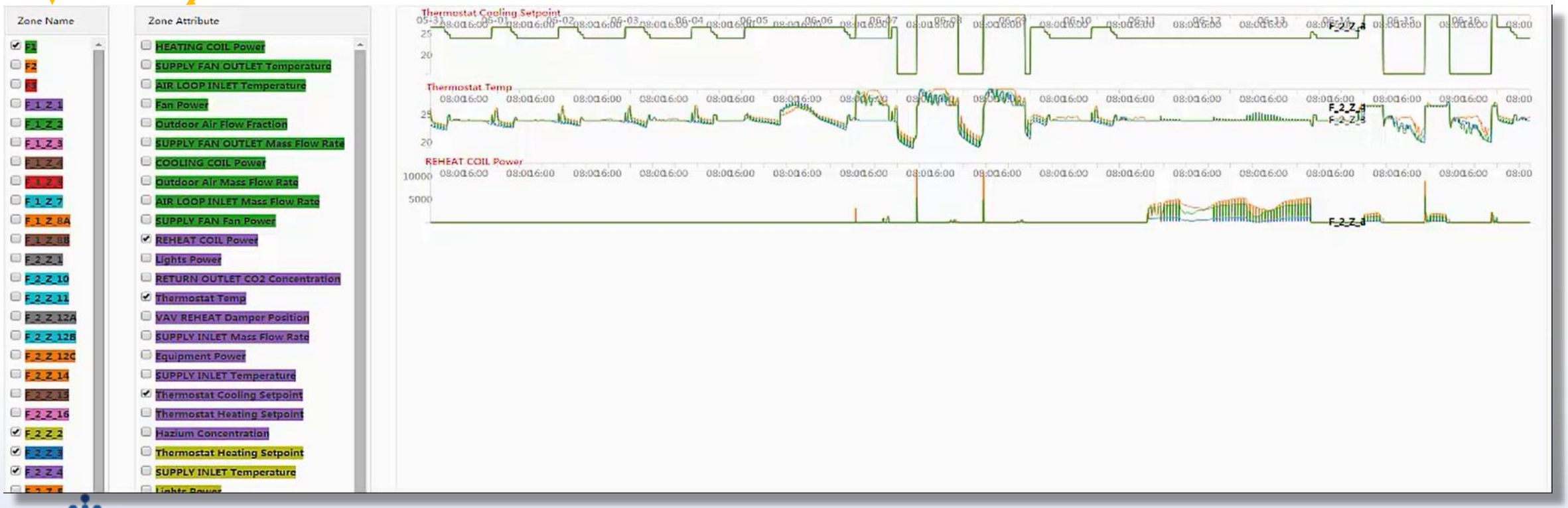
Work flow

- Two systems
  - System A (Labelling System)
    - Basic visualizations + labelling
    - Store insight of the data
  - System B (Analysis System)
    - Exploits the insight from system A to reduce the ambiguity of the tasks
    - Anomaly-detection as entrance

System A: Time Series Interface

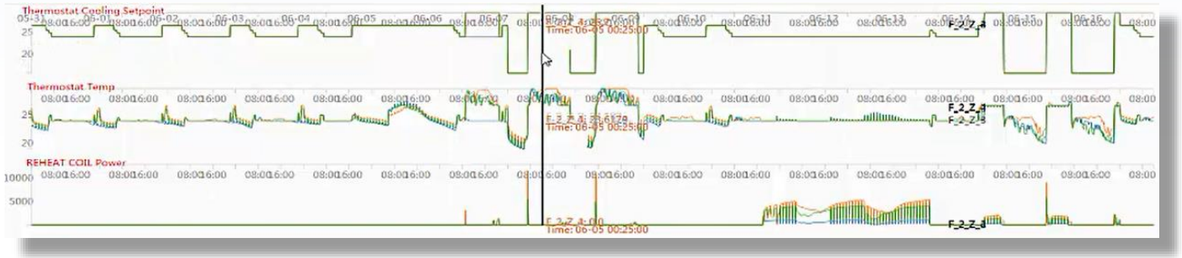
Spatial filter

Attribute filter



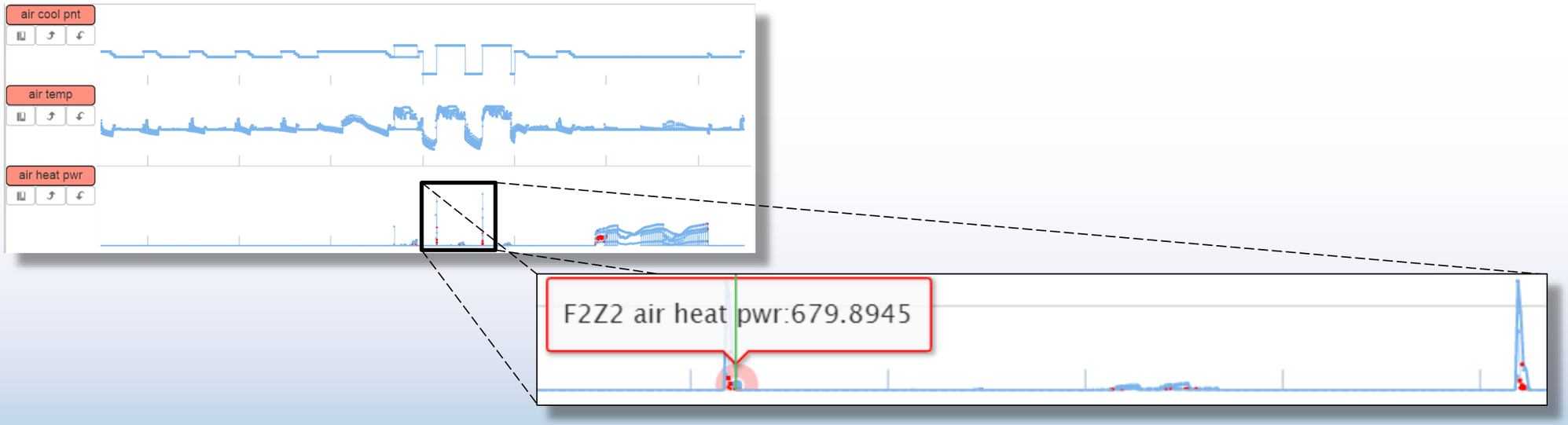
Time Series Interface

System A



Insight 1: Curves in weekdays differs from that in weekends
Insight 2: Periodical pattern in weekdays
Anomaly detection metric: template of a weekday (show anomaly)

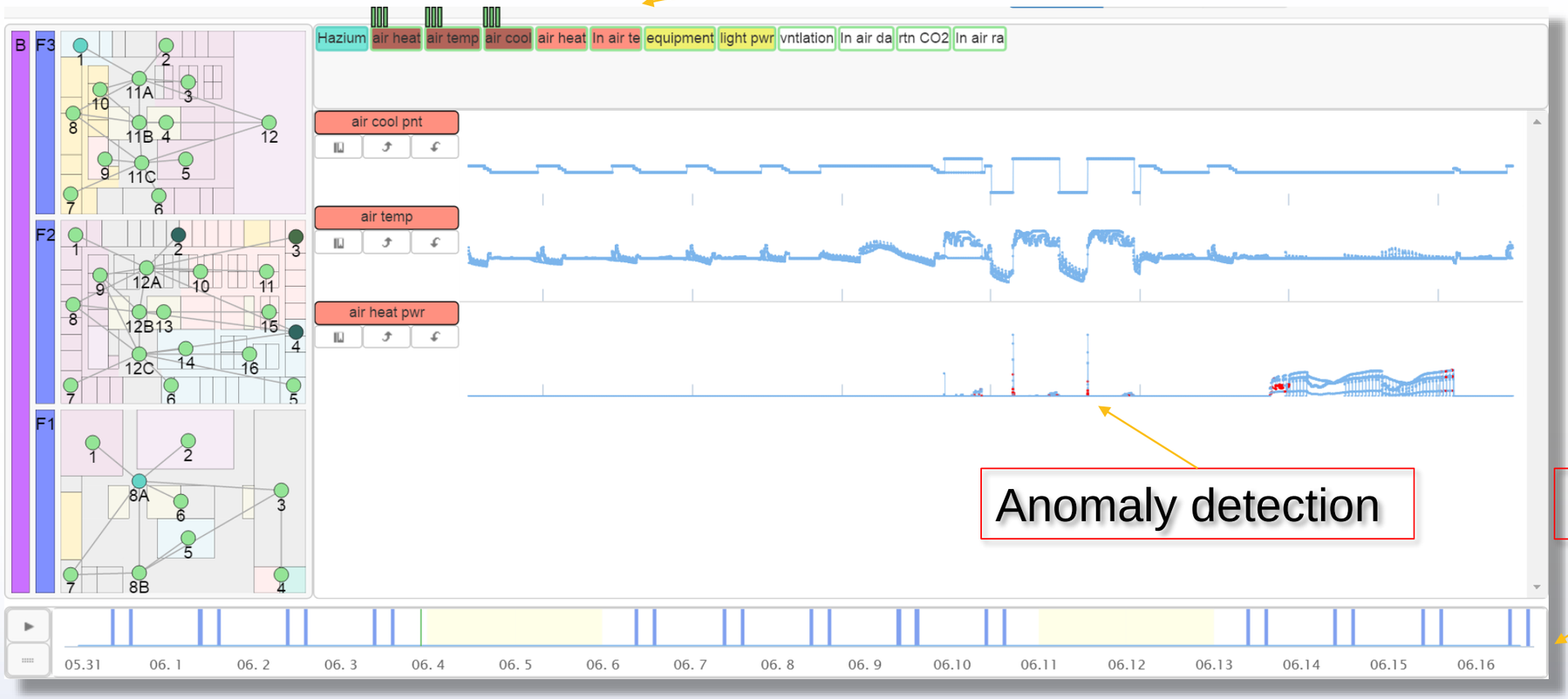
System B



System B: Time Series Interface

Spatial filter

Attribute filter

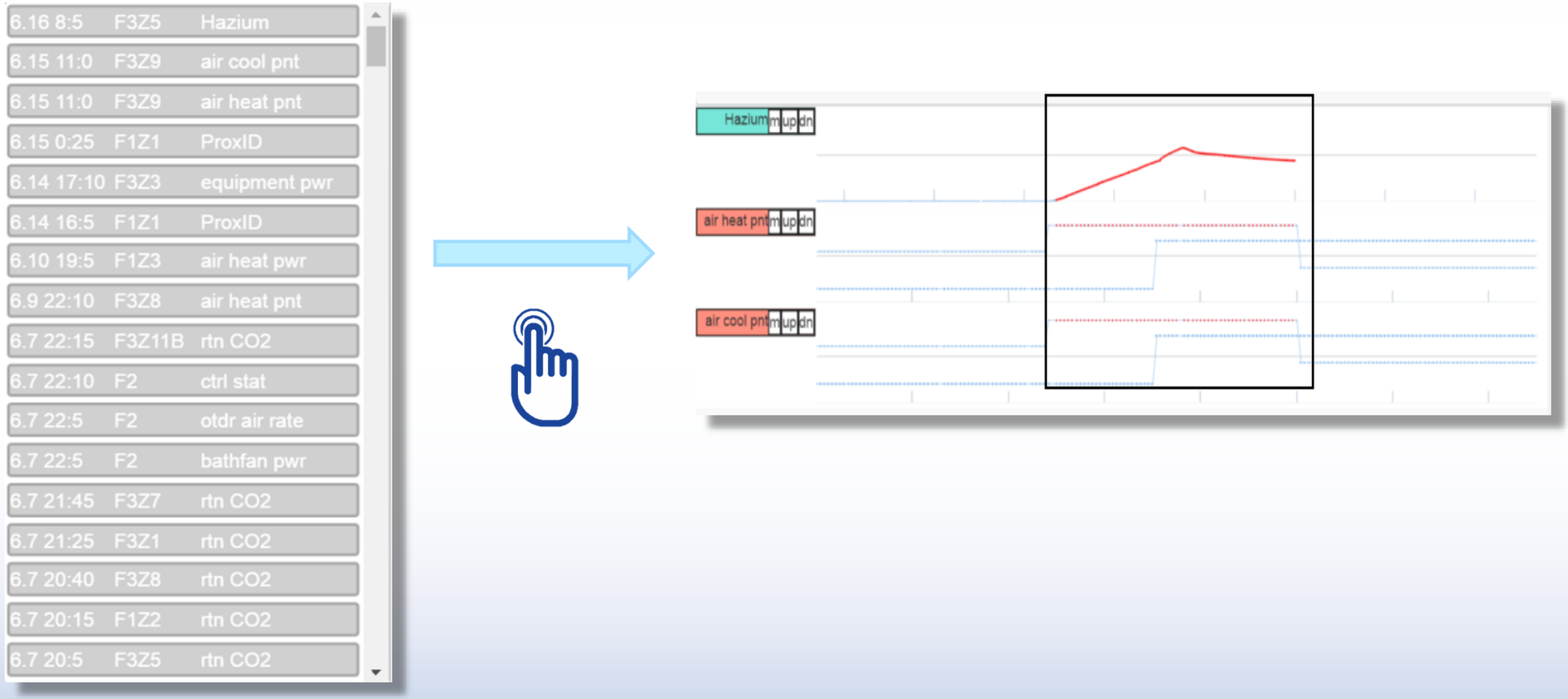


Anomaly detection

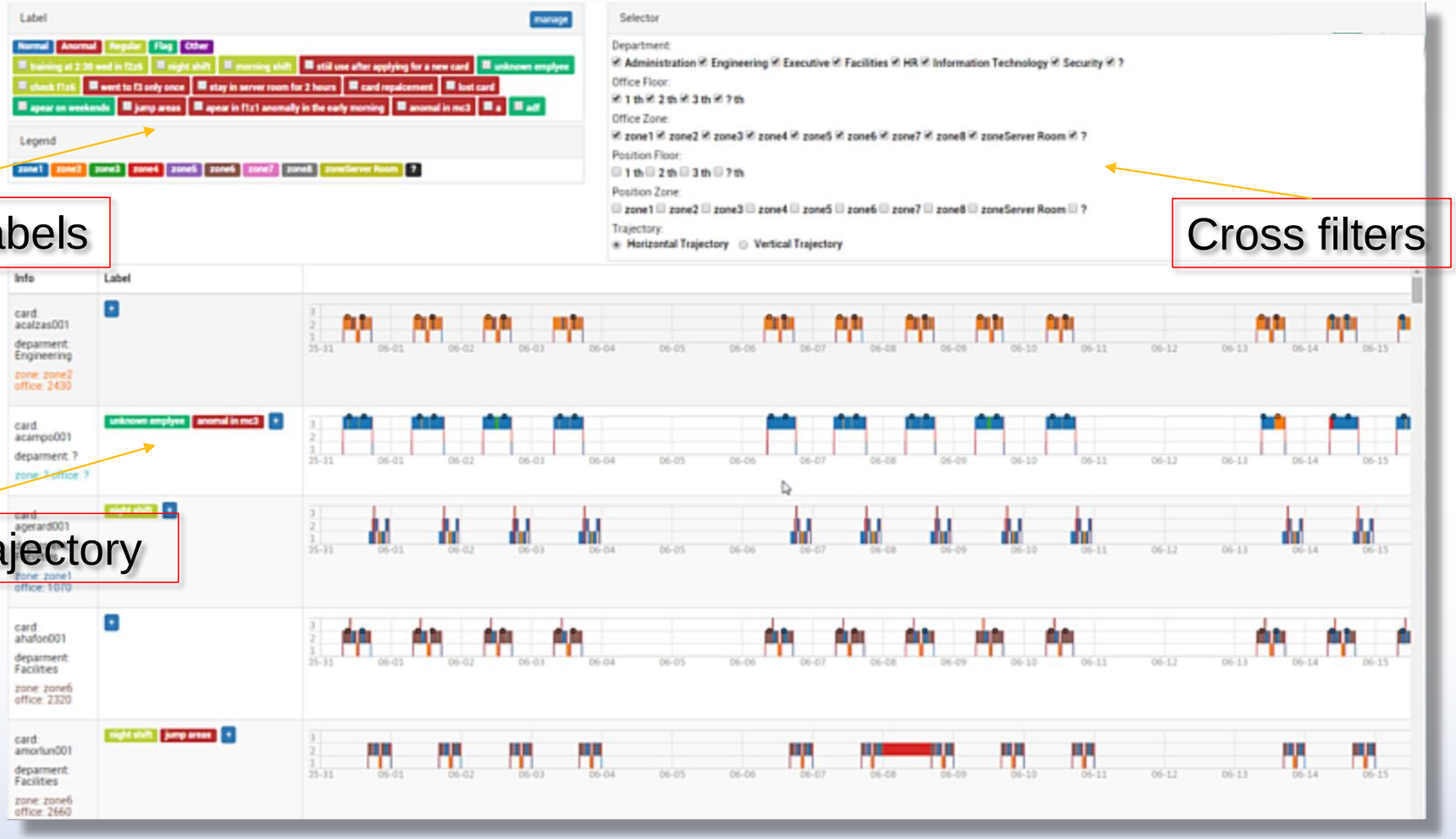
Temporal filter

System B: Warning Stack

- Store all the anomalies with the metadata <time, position, attribute>



System A: Trajectory Interface



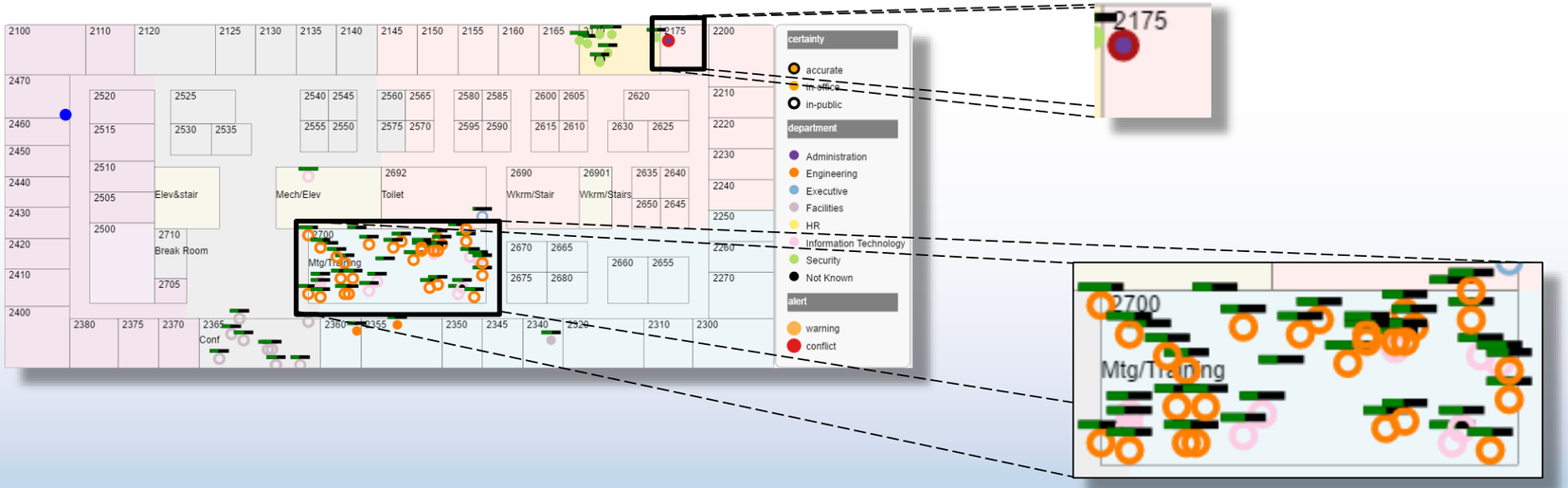
Trajectory Interface

System A



Insight 1: Physically impossible events / suspicious events
Design requirement 1: Give warnings (show anomaly)
Insight 2: Meeting events
Design requirement 2: Directly visualize the meeting (show pattern)

System B



Trajectory Interface

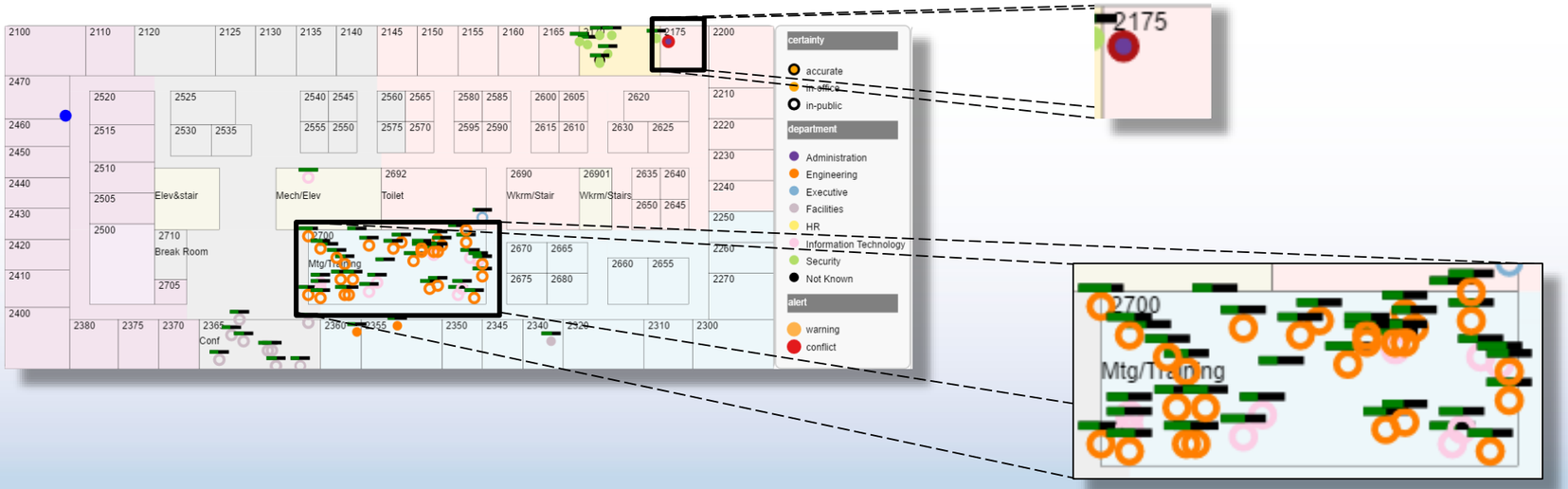
System A



Anomaly detection metric:
1. Move between zones that are not adjacent - Strong
2. Conflict between two trajectory data source - Strong
3. Staying in a zone that contains neither the office of the employee nor public area - Weak

Insight 1: Physically impossible events / suspicious events
Design requirement 1: Give warnings (show anomaly)
Insight 2: Meeting events
Design requirement 2: Directly visualize the meeting (show pattern)

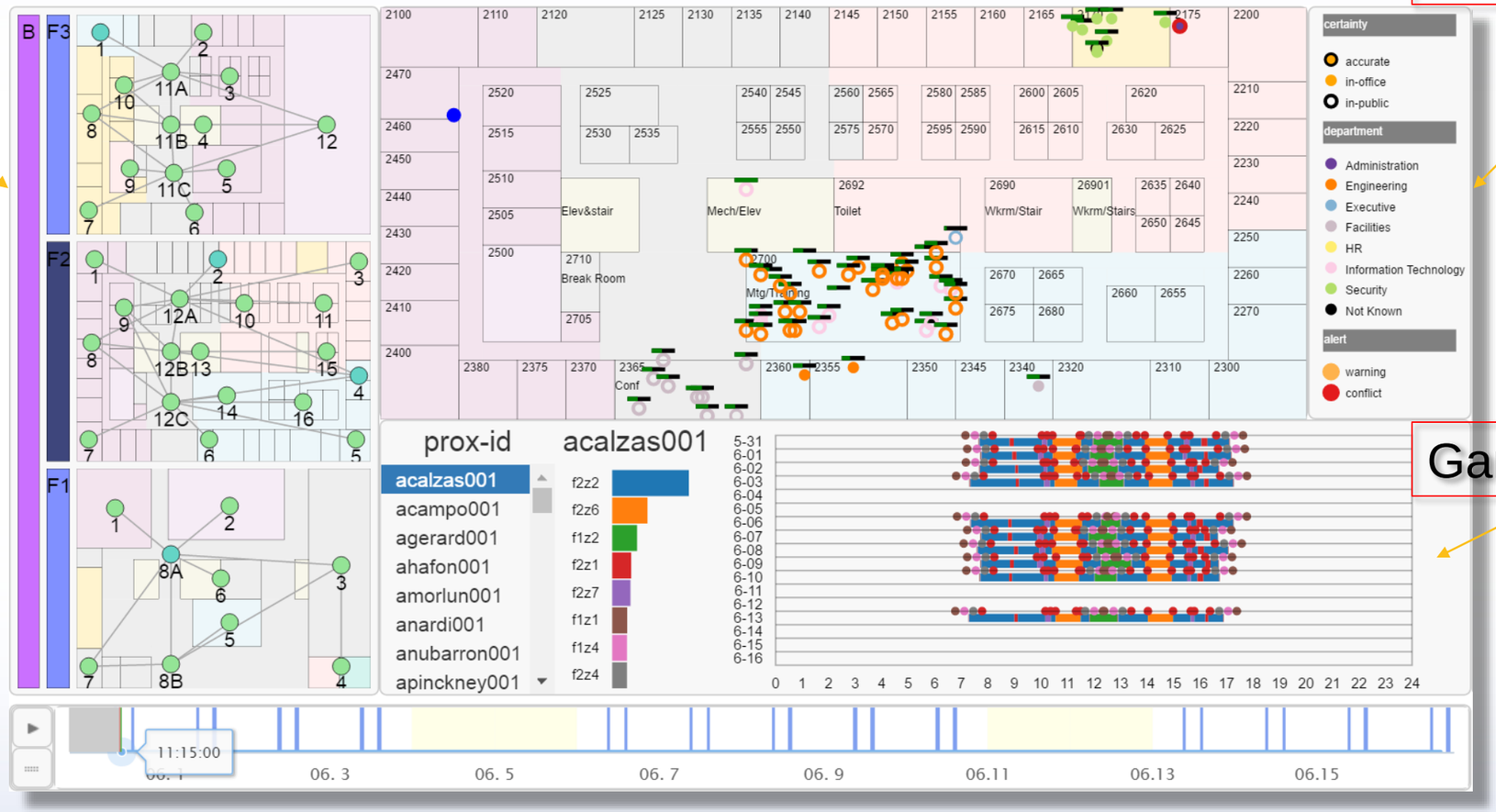
System B



System B: Trajectory Interface

Spatial filter

Trajectory simulation



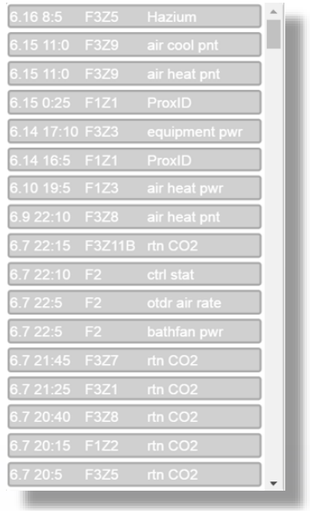
Gantt chart

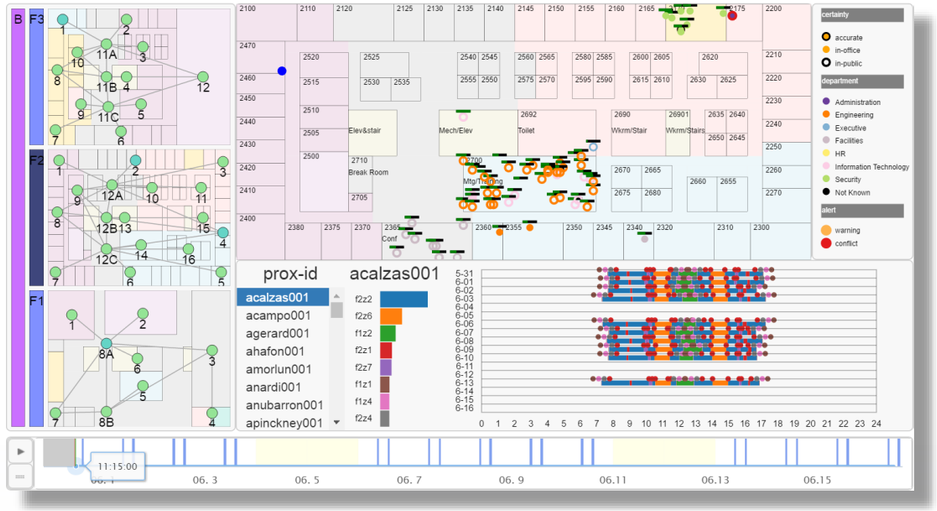
System B: Trajectory Interface

- Data source
  - Fixed prox sensor - spatial uncertainty
  - Mobile prox sensor - temporal uncertainty
- Uncertainty reduction
  - Robot detection
  - Stay in office
  - Stay in public areas (e.g. meeting room)
  - Stay in the corridor

System B: Streaming Data Interface







System B: Streaming Data Interface

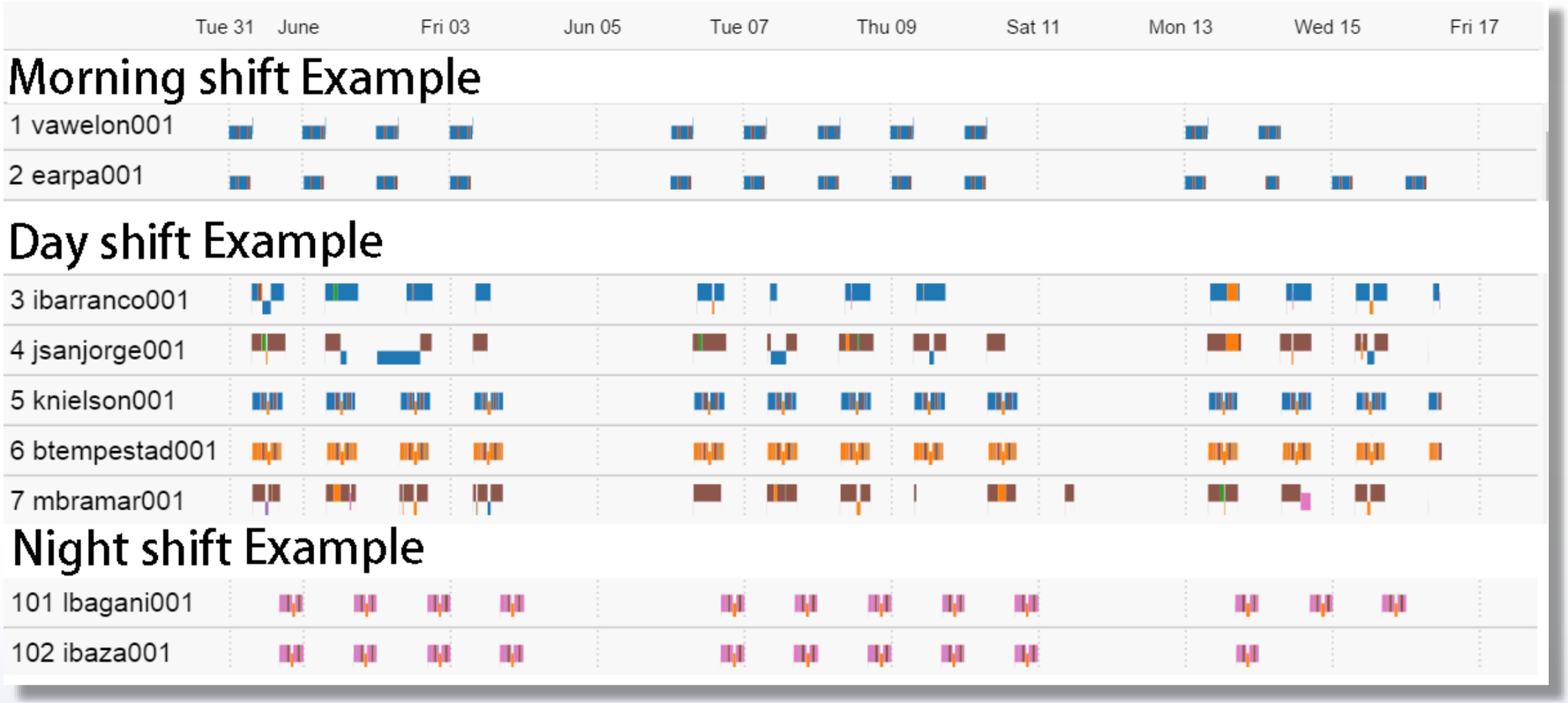
Position-centric glyphs

Attribute-centric glyphs

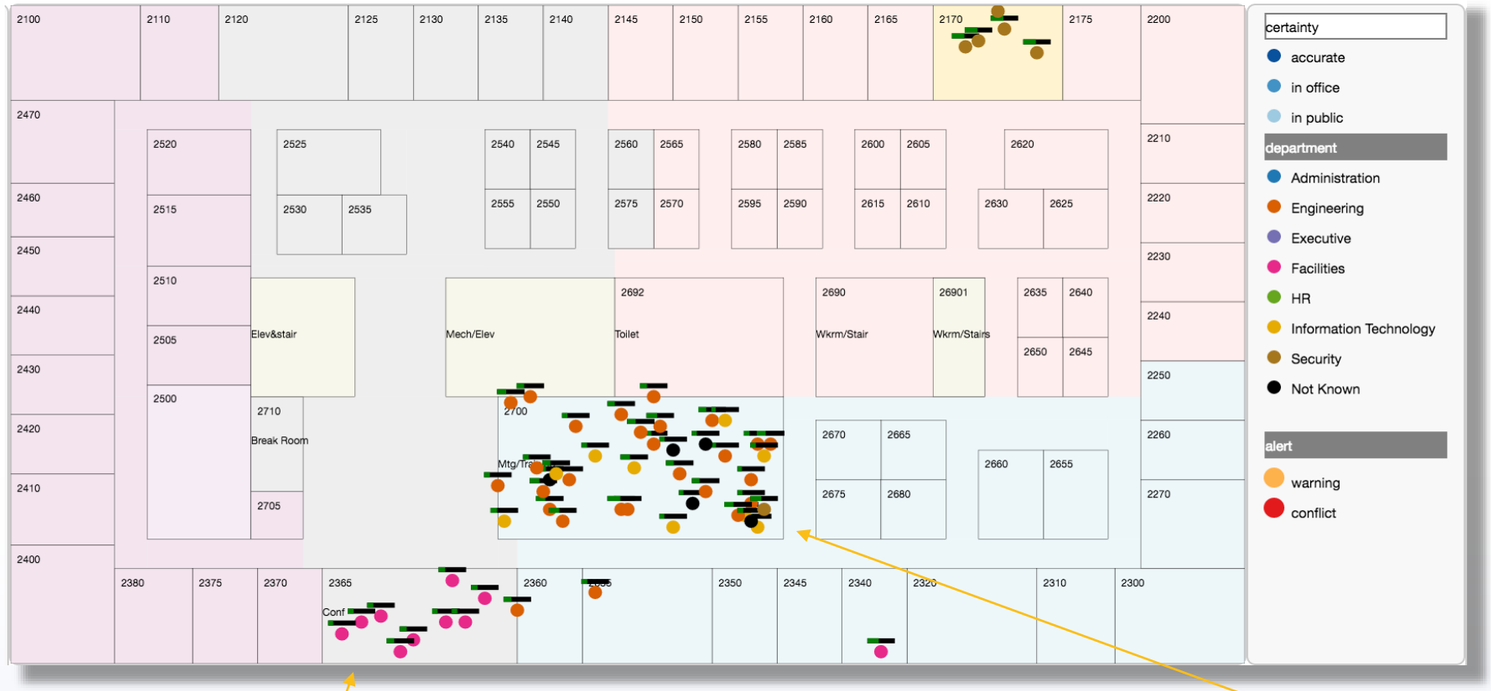
Gantt Chart



Case 1: Shifts



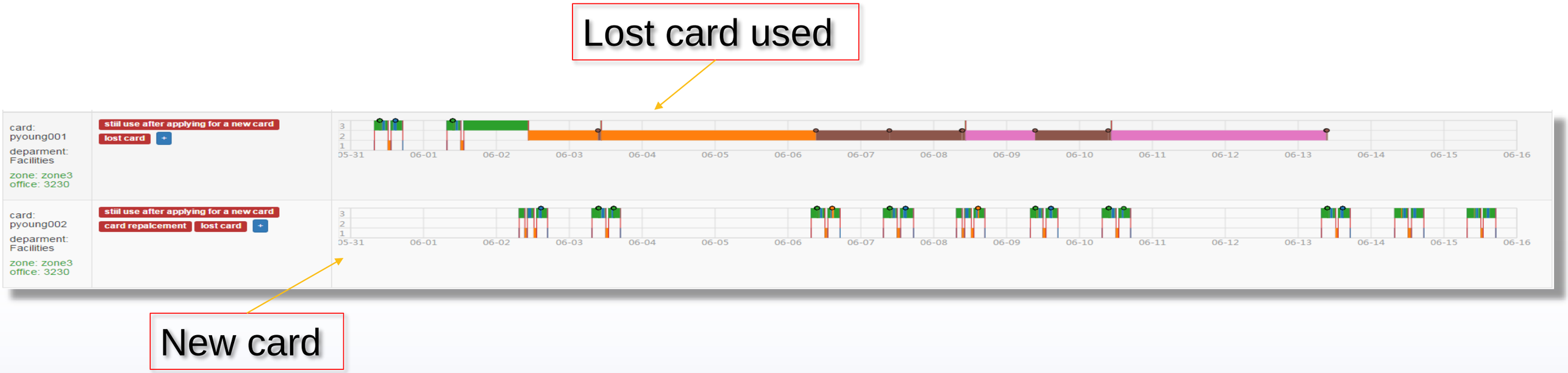
Case 2: Meetings



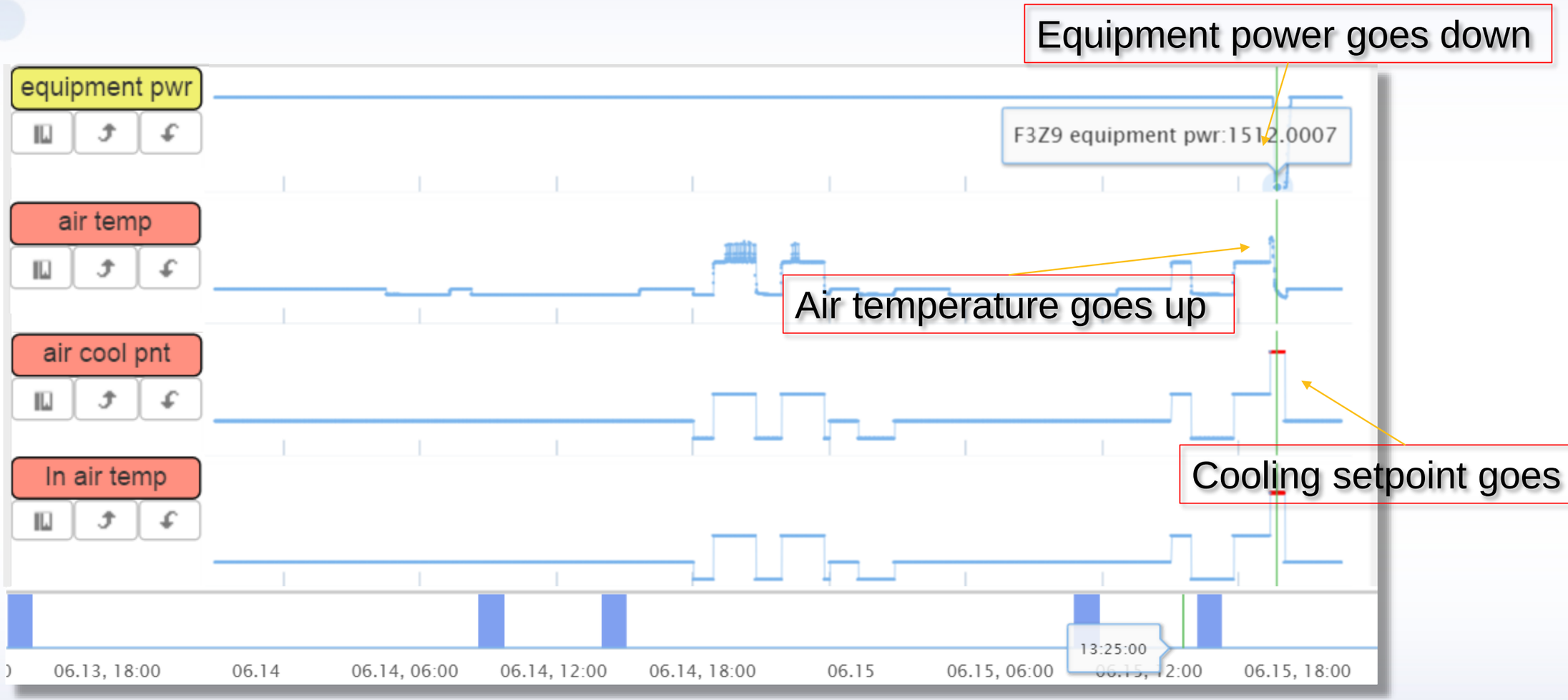
Information technology and engineering meet at Mtg/Training (2700)

Facilities meet at Conf (2365)

Case 3: Lost card



Case 4: Server Room down time



Conclusion

- STAD-HD: twin systems for specifying and visualizing patterns and anomalies in heterogeneous dataset
  - Labelling System for insight + Analysis System for answers
- Drawbacks
  - Labelling System cannot automatically generate insights
  - Transformation from insight to design requirements and finally to implementation is not automatic
  - Analysis System reports many false positives

Acknowledgement

- Funding
  - NSFC No. 61170204
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- Reviewers
  - Anonymous Reviewers



Thank You

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